Day 31 - [8.3.1] Polynomial Division

Wednesday, May 8, 2024 10:46 AM

Objective:	**Key Notes:**

• Students will divide polynomials using area models.
 	
Mathematical Practice(s): reason abstractly and quantitatively, attend to precision, look for and make use of structure
 	
Classwork:
• Polynomial Division
 	
Homework:
• Day 31 HW
 	
Announcement:
• Day 33: Polynomial Team Task
  ○ ODD - Monday, 4/28
  ○ EVEN - Tuesday, 4/29
• Day 34: KC #1 - Polynomials
  ○ ODD - Wednesday, 4/30
  ○ EVEN - Thursday, 5/1	

BELLringer

Fill in the following table.

	1	2	3	4	5
A	$\times$	$4x^3$	$+6x^2$	$-2x$	-5
B	$2x$				
C	-3				

- **Classwork**

  8-115. POLYDOKU TEAM CHALLENGE

  Work with your team to complete the puzzle below. Find the factors and the product for the puzzle. If you get stuck, you can consult parts (a) through (c) below for ideas.

	1	2	3	4	5
A	$\times$			$-2x$	
B	x	$2x^4$			
C	-4		$12x^2$		
				$12x$	

a. How is cell B2 related to the answer?

b. How did you find the third term in the answer?

c. What cells did you use to get the value in cell B5?

8-116. Jessica is about to start the intermediate-level Polydoku puzzle shown below. Show Jessica how to complete the puzzle. Make sure you can justify your solution.

	1	2	3	4
A	$\times$			
B	$2x$			
C	$+5$			
	$\mathbf{6x^3}$	$\mathbf{+7x^2}$	$\mathbf{-16x}$	$\mathbf{+10}$

Use your results to complete the statements below.

$\frac{6x^3+7x^2-16x+10}{2x+5}=$ _____ , THEREFORE

$(2x+5)\cdot$ _____ $=$ _____

8-118.
Create your own Polydoku puzzles that can be used to solve each of the polynomial-division problems below. Express any remainders as fractions and use your results to write a multiplication and a division statement such as those in problem 8-117.

a. $\dfrac{6x^4-5x^3+10x^2-18x+5}{3x-1}$

b. $\left(x^4-6x^3+18x-4\right)\div(x-2)$

c. $\left(x^3+x^2-14x+3\right)\div(x-3)$

d. $\dfrac{x^5-1}{x-1}$

What does your work above tell you about the roots of the polynomials?
Graph them on Desmos and look for a pattern.

METHODS AND MEANINGS

MATH NOTES

Polynomial Division

The examples below show two methods for dividing $x^4 - 6x^3 + 18x - 1$ by $x - 2$. In both cases, the remainder is written as a fraction.

Using long division:

$$
\begin{array}{r}
x^3 - 4x^2 - 8x + 2 \\
x-2 \overline{\big) x^4 - 6x^3 + 0x^2 + 18x - 1} \\
\underline{x^4 - 2x^3} \\
-4x^3 + 0x^2 \\
\underline{-4x^3 + 8x^2} \\
-8x^2 + 18x \\
\underline{-8x^2 + 16x} \\
2x - 1 \\
\underline{2x - 4} \\
3
\end{array}
$$

Using an area model:

	x^3	$-4x^2$	$-8x$	$+2$	Remainder
x	x^4	$-4x^3$	$-8x^2$	$+2x$	3
-2	$-2x^3$	$+8x^2$	$+16x$	-4	
	$x^4 - 6x^3$	$+0x^2$	$+18x$	-1	

Answer: $x^3 - 4x^2 - 8x + 2 + \frac{3}{x-2}$

Answer: $x^3 - 4x^2 - 8x + 2 + \frac{3}{x-2}$

Therefore, $\left(x^4 - 6x^3 + 18x - 1\right) \div (x - 2) = x^3 - 4x^2 - 8x + 2 + \frac{3}{x-2}$ and $(x-2)(x^3 - 4x^2 - 8x + 2 + \frac{3}{x-2}) = x^4 - 6x^3 + 18x - 1$.

Answers can also be written in factored form with a remainder. For example, the answer above would be written as

$x^4 - 6x^3 + 18x - 1 = (x - 2)\left(x^3 - 4x^2 - 8x + 2\right) + 3.$